

Recursive Genomic Q-Former: Parameter-Reduction vs. Performance Trade-off

Task details	
Task	cancer_type — 4-way tumor-of-origin classification from bulk gene-expression profiles (TCGA RNA-seq).
Dataset	Unified TCGA cohort: 2,738 samples $\times$ 20,530 genes; top 4,000 highly-variable genes used as input tokens.
Classes	Breast (BRCA, 867), Head-neck (HNSC, 423), Lung (LUNG, 943), Prostate (PRAD, 505).
Split	70/15/15 train/val/test, 523 test samples , seed 42.
Model	Recursive Genomic Q-Former: $d_{\text{model}} = 128$, 256 marker query slots, recursion depth 4.
Metric	Test accuracy and macro-averaged F1.

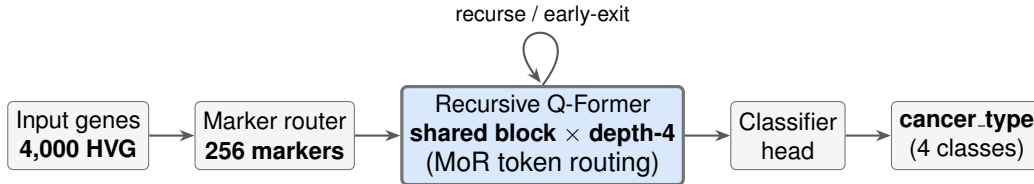


Figure 1: Pipeline. The shared transformer block (blue) is reused across recursion steps; MoR routes tokens to a per-token depth instead of a fixed pass.

All results are on the **cancer_type** classification head (4 classes: BRCA, HNSC, LUNG, PRAD; 523 test samples), with a fixed recursion depth of 4, $d_{\text{model}} = 128$, 256 marker query slots, and seed 42. The “regular” recursive model shares one transformer block across all recursion steps; the comparison models either un-tie those weights (*Independent*) or add token-adaptive routing (*MoR*).

Table 1: Main parameter-reduction trade-off. Sharing the transformer block across the 4 recursion steps cuts transformer parameters by $\sim 4\times$ versus an un-shared (independent) stack, while losing only 0.57 accuracy points. Mixture-of-R recursions (MoR) token routing keeps the shared (small) parameter count yet recovers most of the gap and saves $\sim 41\%$ compute.

Model	Weights	Routing	Transf. params	Param ratio	Test acc.	Macro-F1
Independent stack	untied	none	530,432	$3.99\times$	0.9904	0.9891
Recursive (shared)	shared	none	132,992	$1.00\times$	0.9847	0.9871
MoR (token-adaptive)	shared	token	132,992	$1.00\times$	0.9885	0.9892

Read-out. The shared recursive model uses **74.9% fewer** transformer parameters than the independent stack (132,992 vs. 530,432) for a **0.57-point** accuracy cost. Adding MoR token routing

closes that gap to **0.19 points** at the same small parameter budget, and is the only configuration that also reduces FLOPs.

Table 2: Parameter scaling vs. recursion depth. Shared-weight recursion keeps the transformer parameter count essentially flat as depth grows, whereas an independent stack grows linearly. The reduction factor reaches nearly 8 $\times$ at depth 8.

Recursion depth	Shared params	Independent params	Reduction factor
1	132,608	132,608	1.00 $\times$
2	132,736	265,216	2.00 $\times$
4	132,992	530,432	3.99 $\times$
6	133,248	795,648	5.97 $\times$
8	133,504	1,060,864	7.95 $\times$

Table 3: Compute trade-off (depth-4 variants). Early-exit / adaptive recursion lowers the *effective* mean depth and FLOPs per sample relative to a fixed depth-4 pass, at equal parameter count. MoR achieves the lowest compute and the best F1.

Configuration	Transf. params	Mean depth	MFLOPs/sample	Test acc.
Fixed depth-4 (no early exit)	132,480	4.00	268.4	0.9847
Recursive (shared, early-exit)	132,992	2.75	161.5	0.9847
No recursive refine	132,992	2.75	161.5	0.9771
MoR (token-adaptive)	132,992	2.69	159.1	0.9885

Notes. Param ratio in Table 1 is independent/shared transformer parameters. “Transf. params” counts only the recurrent transformer block (the full model incl. embeddings/heads is $\sim 712\text{K}$ for shared variants, $\sim 1.11\text{M}$ for the independent stack). Compute-saving for the early-exit and MoR variants is $\approx 40\%$ of the fixed depth-4 FLOP budget. Token-reduction at the marker stage additionally compresses 4000 input genes to 256 retained markers (93.6% token reduction) at 98.5% accuracy.